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Master's Degree Program - Specializations:
Artificial Intelligence, Data Science, Networks and Distributed Systems

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Master : Artificial Intelligence, Data Science, Networks and Distributed Systems
Subject : Advanced Algorithms and Complexity

Tutorial Sheet - TD 1

Algorithmic Complexity and Asymptotic Notation

Corresponds to: Lectures 1-2

Learning Objectives

- Define and use the O, Ω and Θ notations correctly.
- Classify standard growth-rate functions from slowest to fastest.
- Compute the tight complexity of simple loop-based pseudocode.

Exercise 1

Order the following functions from slowest to fastest asymptotic growth, and justify your ordering (you may use limits of ratios or known growth hierarchies):

$n!,\ 2^n,\ n^3,\ n \cdot \log n,\ \log n,\ n^2,\ \sqrt{n},\ n$

Exercise 2

Give the tight bound **Θ(·)** for the number of elementary operations performed by the following pseudocode, as a function of **n**:

```
count = 0
for i = 1 to n:
    for j = i to n:
        count = count + 1
```

Exercise 3

Give the tight bound **Θ(·)** for the following pseudocode, and justify each factor of your answer separately:

```
for i = 1 to n:
    j = 1
    while j < n:
        j = j * 2
        do O(1) work
```